ecbench round 2 run report: walk key ring, 32-bit inversion, raw-clock IC trace

Run start: 2026-11-01T20:10Z (UTC). **Study:** `research/ecbench_walk_trace_ir_20261101/` (README with the tables and the reasoning). **Class:** engineering; every child report identical apart from timing fields.

What ran

- **Binaries.** E: 3144 head (sha256 0038806674a3af21…); F: E + `TunedWalker` key ring, jump wrap without division, 32-bit `invmod` (b6cd58faf502e51a…); G: F + the IC phase trace on raw `CLOCK_MONOTONIC` nanoseconds (1efe6b068214f08e…).
- **Inputs.** The exported 32-bit child inputs (`data/input-32bit-*.json`, byte-identical under every binary) and the 20-bit `audit-ir` seq 0–2 of `spec-p20-profile.json`.
- **Instruments.** Callgrind (`--cache-sim=no --branch-sim=no`): the marker split for the rho child, the whole process (`Collected`) for the IC child (its per-relation dump parts are discarded), `audit-ir` at 20 bits. Wall: the child's own `solve_wall_ns`, interleaved rounds E / G / E, `taskset -c 3`, host load about 1.
- **Host.** Shared 4-vCPU cloud VM, Intel Xeon @ 2.10 GHz, no PMU; L0. valgrind 3.22.0, rustc 1.97.0.
- **Completed.** 3 arms × (rho marker run, rho whole, IC whole) at 32 bits; 3 arms × 3 `audit-ir` children at 20 bits; 9 + 7 wall rounds. No failures. `verify --replay-all --exit-code` on the 36- and 4-record scratch sessions under F and G: all reproduced. `cargo test` (ic_boundary, workload, curve_id, ic_measurement): 77 passed, 0 failed; clippy clean apart from the pre-existing unknown-lint warning.

Measured values

Callgrind Ir	E	F	G	E → G
32-bit `rho.negation` solve (34,156 GAE)	30,092,881	24,782,352	**24,781,851**	−17.6%
32-bit `rho.negation` whole child	32,598,131	27,330,279	**27,318,553**	−16.2%
32-bit `ic.pipeline` N=1024 whole child	916,254,438	845,735,563	**776,313,670**	−15.3%
20-bit `rho.negation` solve (1,706 GAE)	1,222,979	1,054,933	**1,054,467**	−13.8%
20-bit `ic.pipeline` N=32 solve (14,101 GAE)	15,144,353	14,373,889	**12,409,320**	−18.1%
20-bit `ic.pipeline` N=45 solve (10,632 GAE)	10,267,488	9,737,258	**8,551,931**	−16.7%

Wall (median, descriptive): 32-bit `rho.negation` solve 6.57 ms → 5.84 ms (A/A 6.51 ms, 9 rounds); 32-bit `ic.pipeline` solve 278.6 ms → 261.7 ms (A/A 279.6 ms, 7 rounds); round-to-round spread ±10–15%.

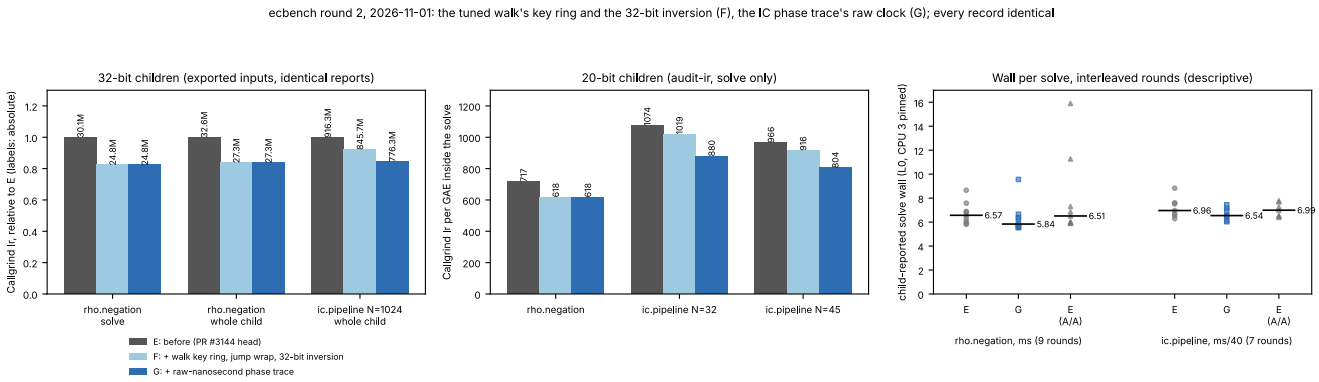


Figure 1: Recorded values only (`data/ir_32bit.tsv`, `data/ir_20bit.tsv`, `data/wall_rounds_32bit.tsv`). Left: 32-bit Ir relative to E with absolute labels. Middle: 20-bit solve Ir per GAE. Right: every wall round with its median.

Raw records

`data/ir_32bit.tsv`, `data/ir_20bit.tsv`, `data/wall_rounds_32bit.tsv`, `data/input-32bit-rho-negation.json`, `data/input-32bit-ic-pipeline.json`. Scratch sessions as in `research/ecbench_child_walk_ir_20261101/` (not committed).